

QUIZ 9-B (Spring-2024)

ID: _____ NAME: _____ SECTION: _____

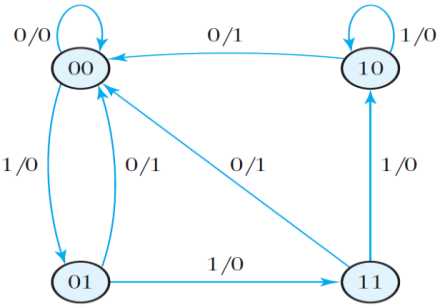
1. Design the sequential circuit using **T-Flip-flop** following given circular sequence **0->2->3** back to 0 where x is control input for x=0 sequence counts **UP** whereas for x=1 sequence counts **down**? find self-correcting state **1**? You are required to produce state table and state equations.

Self-Correction state 0 is for x=0 _____ and for x=1 _____

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2. For the following state diagram, tabulate **reduce state table**. Start from state **L**, and input sequence **01110011**, determine output sequence.
NOTE: Where equivalent state **00=L**, **01=M**, **10=N** and state **11= O**



3. Design **4-bit-Universal Shift Register**? Perform following operations.

S ₂	S ₁	S ₀	Register Operation	S ₂	S ₁	S ₀	Register Operation
0	0	0	Clear to 1	1	0	0	Old State
0	0	1	Complement of Parallel load	1	0	1	Shift Right
0	1	0	Shift Left	1	1	0	Parallel load
0	1	1	No Change	1	1	1	Clear to 0